

Instructions in preparing an application

Applicants are required to use this template to submit their Flagship program. Minimum font 11pts (Arial, Calibri, Times New Roman), standard line and character spacing, and normal margins. All indicated chapters are mandatory unless otherwise indicated. You must respect the indications given in each chapter in order to adhere to the expected content. A word count is clearly indicated for each section. **If you do not adhere to the maximum amount of words, your application will not be processed.** The word count includes tables, footnotes and figure legends but excludes references. Please do not include hyperlinks. The application form needs to be submitted in a single pdf document, including the budget table and letters of intent. We recommend that you do not send in your submission at the last minute in order to avoid technical difficulties. No submission will be accepted after the deadline, March 15th at 13:00 CET. Applications should be sent in PDF format to ***Flagships@erasmusmc.nl***.



1. Title

CODE RED: Convergence Dedicated to the Reduction of Expenditure on Drugs



General information

1.a Applicants

In case an applicant has more than one affiliation, please mention this in the table.

Name*	Institution	Department	Email	Tenured (i.e. fixed) position/ Tenure-track/ Other (please specify)	For Erasmus MC applicants: clinician/ researcher/ mix	Role in Flagship [Main Lead/Lead/ Applicant]
Prof. dr. Hugo van der Kuy (Flagship lead from Erasmus MC)	Erasmus MC	Hospital pharmacy	h.vanderkuy@erasmusmc.nl	professor	Mix	Main lead
Prof. dr. BCP (Birgit) Koch	Erasmus MC	Hospital pharmacy	b.koch@erasmusmc.nl	professor	Mix	Work package lead
Prof. dr. Carin Uyl- de Groot (Flagship lead from Erasmus University Rotterdam)	EUR	HTA, Erasmus School of Health Policy and Management	Uyl@eshpm.eur.nl	professor	Mix	Main lead, Work package lead
Eline Bunnik PhD	Erasmus MC	Department of Medical Ethics, Philosophy and History of Medicine)	e.bunnik@erasmusmc.nl	Associate Professor	Researcher	Work package lead
Prof. dr. J.J. (Jan) Cornelissen	Erasmus MC	Department of Hematology	J.cornelissen@erasmusmc.nl	professor	Researcher and Chair Medical Ethical Committee (METC)	Work package lead

	European Medical Agency	Human Medicines, Hematology-Oncology Department	jan.cornelissen@ema.europa.eu	National expert	-	Work package lead
Prof. dr. A.M. (Marileen) Dogterom (Flagship lead from TU Delft)	TU Delft	Department of Bionanoscienc e	m.dogterom@tudelft.nl	Professor	-	Main lead, Work package lead
Jan-Dietert Brugma MSc	Erasmus MC	Outpatient Pharmacy	j.brugma@erasmusmc.nl	Fixed position	Clinician	Work package lead
Christopher Danelon PhD	TU Delft	Department of Bionanoscienc e	c.j.a.danelon@tudelft.nl	Associate Professor	-	Applicant
Asterios Katsifodimos PhD	TU Delft	Department of Software Technology	a.katsifodimos@tudelft.nl	Assistant Professor	-	Applicant
Prof. Dr. Frank W.G. Leebeek	Erasmus MC	Department of Hematology	f.leebeek@erasmusmc.nl	Professor	Mix	Work package lead
Dr. R.W.F. (Roelof) van Leeuwen	Erasmus MC	Departments of Hospital Pharmacy and Oncology	r.w.f.vanleeuwen@erasmusmc.nl	Fixed position	Researcher	Applicant
M.P.J.K. (Martijn) Lolkema, MD PhD	Erasmus MC	Department of Medical Oncology	m.lokema@erasmusmc.nl	Fixed position	Mix	Applicant
Prof. dr. A. (Aad) van der Lugt	Erasmus MC	Department of Radiology	a.vanderlugt@erasmusmc.nl	Professor	Mix	Applicant
Prof. dr. A.H.J. (Ron) Mathijssen	Erasmus MC	Department of Internal Oncology	a.mathijssen@erasmusmc.nl	Professor	Mix	Applicant
Nestor Parolya PhD	TU Delft	Delft Institute of Applied Mathematics	n.parolya@tudelft.nl	Assistant Professor	-	Applicant
Prof. dr. M.H.N. (Maartje) Schermer	Erasmus MC	Department Medical Ethics, Philosophy and History of Medicine	m.schermer@erasmusmc.nl	Professor	Mix	Applicant
Frederick Thielen PhD	EUR	HTA, Erasmus School of	thielen@eshpm.eur.nl	Assistant professor	-	Applicant

		Health Policy and Management				
Rihan Hai	TU Delft	Department of Software Technology	r.hai@tudelft.nl	Assistant professor	-	Applicant

Corresponding Lead: Hugo van der Kuy

Planned start date of seed funding: 01/07/2022

Primary Convergence theme: *Improving Health Journeys*

Secondary Convergence theme (if applicable): *Technology Supported Transitions in Health(care)*

Tertiary Convergence theme (if applicable): *Human Centered Technology and AI for Health*

Quaternary Convergence theme (if applicable): *Fundamentals of Health & Disease*

Keywords: *drug expenditure, sustainable healthcare system, patient access, personalized treatment, AI, cost-effectiveness, patient engagement*

1.b Summary (max 300 words)

Aim

To structurally reduce pharmaceutical expenditure across the drug lifecycle and to improve patient care using innovative technology and knowledge convergence.

Objectives

To:

- I. Develop drug cost reduction frameworks for sustainable health care systems.
- II. Gain knowledge and stimulate the application of personalised treatment for patients to improve treatment outcomes and to optimize precision drug dosing.
- III. Promote a data-driven approach in which human-centred artificial intelligence (AI) and machine learning (ML) are applied as tools to ensure transferability of findings
- IV. Establish innovative methods to accelerate registration and stimulate local production of innovative drugs.
- V. Create a research environment that stimulates convergence between the different stakeholders and disciplines across the drug lifecycle.
- VI. Ensure long-term sustainability of the knowledge and tools produced by **CODE RED** by impact evaluation and stakeholder engagement throughout and beyond the project.

Design

To control expenditure on drugs, all phases of the drug lifecycle should be (re)assessed. We aim to launch seven work packages that help to address a comprehensive approach to drug pricing and expenditure while improving quality of care for patients and ensure patient access:

- Academic innovation
- Local production & registration
- Alternative pricing and reimbursement
- Purchasing methods
- Societal & patient engagement
- Sustainable & precision dosing
- Real-world outcomes & cost-effectiveness

Anticipated impactScientific

Producing new knowledge, models and methods that potentially reduces costs of expensive and innovative drugs throughout their lifecycle without compromising quality of care. This includes both 'know-what' and 'know-how' on drug research & development, commercialisation, logistics, patient care and sustainability.

Societal

Developing novel, sustainable frameworks and practices that are ethically acceptable, socially accepted and applicable to hospitals, pharmaceutical industry and payers to help reduce costs of pharmaceuticals, improve patient care, and ensure sustainable and equitable patient access to medicines.

Economic

Accomplishing cost containment of (expensive) drugs, for Erasmus MC and for other institutes and stakeholders, nationally and internationally.

1.c Public Summary (max 300 words)**Aim and objectives**

CODE RED aims to reduce the costs of drugs while improving care for patients by reaching more consensus on the right technologies and knowledge to use for the development, commercialisation and application of drugs. Moreover, **CODE RED** will attempt to maintain healthcare affordable, personalise treatment for patients, use artificial intelligence and machine learning to improve quality of care, enable more precise dosing of drugs and subsequent avoidance of spillage, and find ways to accelerate the registration of innovative drugs. Further, **CODE RED** will stimulate collaboration between the different stakeholders to enable better research outcomes, and make certain that the knowledge and tools created by **CODE RED** will be translated to practice sustainably.

Design

CODE RED addresses seven research themes that will each consist of several studies. This way, all relevant aspects of drug innovation and expenses are considered and the quality of care for patients as well as their access to essential drugs can be improved. The seven themes are:

- 1) Academic innovation
- 2) Local production & registration
- 3) Alternative pricing and reimbursement
- 4) Purchasing methods
- 5) Societal & patient engagement
- 6) Sustainable & precision dosing
- 7) Real-world outcomes & cost-effectiveness

Anticipated impact of the project

CODE RED will have a scientific, societal and economic impact. As for the scientific impact, new knowledge, both theoretical and practical, will be created on how to reduce drug costs and improve care for patients. **CODE RED** will have a societal impact by developing new and 'green' ways by which we can scale down drug expenses, improve care for patients and ensure fair access to drugs for patients, now and in the future. Lastly, **CODE RED** will reduce the impact of drugs on

healthcare costs for Erasmus MC directly, followed by other hospitals and public organizations in the Netherlands and Europe.

2.Relevance to Convergence Health & Technology mission

(50%) — max. 2500 words for 2.a to 2.e

2.a Contribution to H&T mission and fit with H&T theme(s)

Contribution to the Health and Technology mission

The rapid growth of pharmaceutical expenditure in the last two decades is increasingly challenging healthcare systems and gaining worldwide attention of policy- and decision makers [1-4]. In 2020, the European Commission adopted the EU Pharmaceutical Strategy to ensure a strong, fair competitive and green industry that delivers for all patients in need [5]. Since orphan drugs and advanced medicinal therapeutic products (ATMPs) such as gene- and cell therapies are increasingly entering the market at soaring prices, new models need to be developed now to safeguard system sustainability and patient access. At the same time there is a need to improve patient care by targeting the right patient group with the right drug at the right dose using personalized medicine.

However, a comprehensive, multidisciplinary approach to address drug expenditure and personalized care across the full drug lifecycle is lacking to date. Therefore, **CODE RED** aims at providing an innovative, multidisciplinary framework to address pharmaceutical expenditure and improve patient care. **CODE RED** adopts a lifecycle-wide approach to foster knowledge and technology convergence across different disciplines (e.g., hospital pharmacy, AI, economics, bio-nanoscience, ethics, computational science, medicine) and institutes across the drug lifecycle. By engaging all stakeholders in a collaborative effort, **CODE RED** initiates system-wide change to address the grand challenges posed by increasing drug expenditure. By establishing a sustainable network in which all partners are engaged and challenged on a continuous basis, effective cross-pollination across disciplines and institutes can be realized. Moreover, **CODE RED** delivers structural contributions to the following ambitions of Convergence for Health and Technology: lifelong health, equality in healthcare, and individualized treatment for all.

Lifelong health

Expensive drugs enter the market for a range of conditions in vastly different patient groups, e.g., rare genetic diseases in children and adults, and types of cancer associated with older age. Hence, the (sometimes extremely) high prices associated with existing and new therapies affect access to essential care in all life phases of patients. **CODE RED** produces *know-what* and *know-how* to reduce expenditure of drugs system-wide and thereby increase patient access to drug treatment to support health from pre-birth to end of life.

Equality in healthcare

Since high drug prices and high budget impact are likely to result in restrictive reimbursement policies, equity of access is increasingly being challenged. Drug expenditures must be curtailed to ensure a sustainable healthcare system and to maintain solidarity within society. **CODE RED** will identify determinants of drug expenditure throughout the drug lifecycle and assess potentially cost-saving innovations by, inter alia, creating synthetic copies of biologics, exploring novel pricing models for pharmaceuticals, and using AI to reach optimal dosing and improve real-world outcomes. Hence, **CODE RED** contributes to improving care and reducing expenditure, which will help uphold our solidarity-based healthcare system.

Individually tailored medical treatment for all

CODE RED further contributes to technology and knowledge convergence by helping to reduce healthcare costs and improving quality of care through personalized drug treatment. Some key examples of this are (see Section 3.c.):

- Developing optimal decision models using AI and ML to reach precision dosing at lower costs. Appropriate data management prior to AI and ML application will ensure the transferability of these models to other settings and institutes and precision dosing models can contribute to more sustainable and safe dosing advice for patients. Recent examples are the more efficient dosing of nivolumab and pembrolizumab.
- Predicting the beneficiary effect of a specific treatment for a particular patient by DNA or RNA analysis of tumor tissue upfront, resulting in a better prediction of treatment outcomes and increased personalized, cost-effective, and sustainable drug prescription.
- Performing pragmatic trials and data modelling based on real-world drug treatment outcomes (including both effectiveness and cost-effectiveness), using AI and ML. This enables the development and subsequent use of optimal decision models in drug treatment. Additionally, these models potentially inform stakeholders involved in drug pricing and reimbursement (e.g., informing performance-based payment negotiations).

Contribution to the Health and Technology themes

CODE RED focuses on all phases of the drug lifecycle to improve quality of drug treatment, enhance affordability and ensure patient access while enhancing ecological sustainability. Moreover, the seven work packages contribute to the Health and Technology programs in the following ways:

Fundamentals of health and disease

CODE RED contributes to the *fundamentals of health and disease* by strengthening the connection between (basic) research and healthcare in practice. To begin with, **CODE RED** explores the potential of creating synthetic copies of biological cells to establish a stable synthetic cell line and (eventually and possibly) decrease the costs of biological drugs. This basic research is potentially revolutionary in drug innovation and more specifically, academic innovation. Further, by applying AI, ML and optimization statistics, tools are developed to guide therapeutic decisions, personalized dosing, and pricing agreements. We will both conduct research as apply this knowledge in field studies within Erasmus MC to help build innovations that are evidence-based transferable.

Improving health journeys

CODE RED helps to improve health journeys both in the short-term and long-term:

- Increasing drug costs may pose a risk to patients throughout their disease, treatment, and recovery through limited access, uncertainty about reimbursement, and having to switch between comparable drugs because of changing substitution policies. **CODE RED** addresses the issue of pricing and reimbursement and subsequent issues through Work Packages 3, 4, 7 (see section 3.c).
- By providing frameworks to engage citizens', patients', and physicians' experiences and preferences regarding several aspects of drugs such as sustainability, pricing and switching. Examples are a citizens panel on fair drug prices and the development of an ethical analysis based on patient' and physicians' preferences in drug switching practices (see Section 3.c., Work Package 5). This way, **CODE RED** helps ensuring timely and affordable access to drugs in accordance with ethical requirements and societal views.
- Regarding the major challenges of climate change and environmental degradation, **CODE RED** considers sustainability as an inherent criterion to be considered when looking at health journeys. Hence, Work Package 6 (see Section 3.c) aims to, inter alia, reduce

pharmaceutical waste by more efficiently dosing and, moreover contributes to less pharmaceuticals being accumulated in the environment.

Human-centered technology & AI for health

Data-driven modelling using classical statistical methods and AI increases the transparency and legitimacy of various decisions across the drug lifecycle. **CODE RED** applies technology and modelling to contribute to evidence-based personalized treatment, as well as innovative pricing and purchasing models. By combining the knowledge gained from modelling, a full understanding of the primary challenges within the drug life cycle and their solution is established. The developed technologies and innovations can be applied in practice across various institutes in the future.

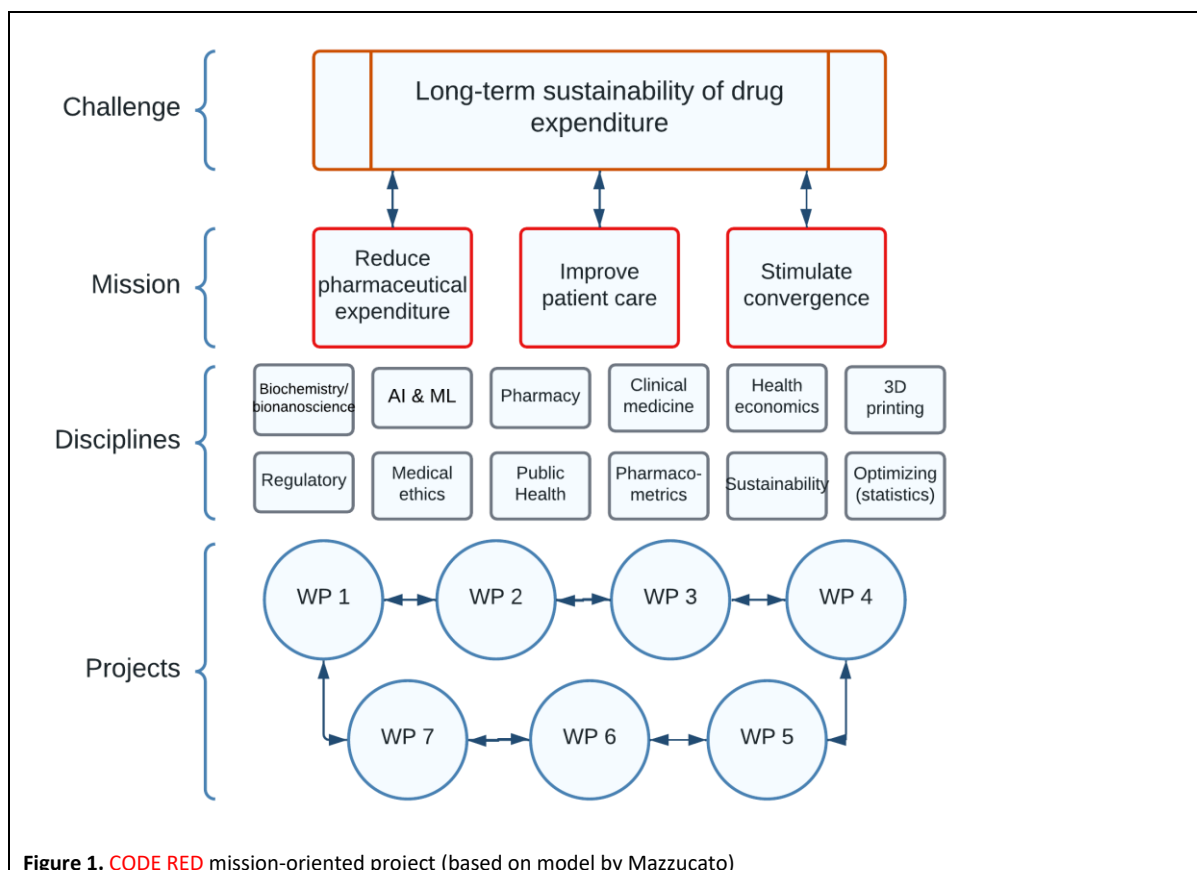
Technology-supported transition in healthcare

Besides conducting research on potential models to reduce costs in the clinical setting, pricing, reimbursement and purchasing models and methods are subject to rethinking. Rethinking models throughout the drug lifecycle with a data-driven approach using innovative technologies in practice enables **CODE RED** to deliver an evidence-based toolset. This toolset can be consulted and applied by various stakeholders throughout the drug lifecycle to safeguard system sustainability and patient access beyond the project itself.

2.b Synergy between disciplines

CODE RED takes on a *mission-oriented approach* to tackle the *grand societal challenge* of drug pricing and expenditure [6] (Figure 1).

Only when EUR, TU Delft and Erasmus MC bundle their capabilities and expertise on pharmacy, computational science, economics, bio-nanoscience, ethics, medicine, and drug procurement, and, moreover, involve external stakeholders, can the dynamic challenges posed by the pharmaceutical market be tackled. More specifically, synergy between the different disciplines is created by; a) bundling different disciplines in each work package; b) establishing a multidisciplinary PhD Work Group and a Student Spirit Team (BA/MA student workgroup) for continuous cross-pollination between young talents; c) holding bimonthly meetings with a multidisciplinary approach involving senior staff, early career researchers and students; and d) creating a multi-disciplinary Advisory Group for the flagship project.



2.c Long term sustainability plan

Long-term sustainability

As our envisioned impact is not only to create structural knowledge but also to develop new frameworks, models, and methods to ensure transferability of results, we need to follow-up on the models developed within **CODE RED**. Therefore, the data-driven innovations developed in **CODE RED** will be assessed on their practical implications through translational research and pilot studies within the five-year project period. Notably, long-term follow up is pivotal to further develop the knowledge and tools produced by **CODE RED** to support long-term sustainability of reduced drug expenditure and to transfer methods to other institutes and stakeholders. Hence, a sustainable research community should be maintained beyond the 5-year period. To realize this, we intend to establish a Convergence drug lifecycle research line chaired by one of the institutions and co-chaired by the other two. Furthermore, we will tighten alliances with other 'academic pharma' initiatives such as NFKC (Leiden), Pharma NL (Dutch *Groeifonds* initiative) and *Medicijn voor de Maatschappij* (Amsterdam).

A potential risk to the long-term sustainability of **CODE RED** is the risk of not maintaining sufficient involvement of the stakeholders needed for coordinated actions to reach the common goal beyond the project duration. **CODE RED** tackles this issue by performing continuous stakeholder engagement and making this a main priority of the Project Coordinator (see Section 3.c.). Together, the Institute Leads, Project Coordinator, and Student Coordinators are responsible for creating a strong and creative academic network through, inter alia, bimonthly meetings, creative communication methods, and adequate progress evaluation (see 2.d.) which are essential for long-term sustainability of relationships.

2.d Visibility, attracting talents and new partners

Visibility

Internally

Visibility of **CODE RED** will be ensured with various tools, captured in a communication plan subject to periodical evaluation of its effectiveness. The establishment of a joint communication plan will be overseen and evaluated by the visibility responsible & stakeholder engagement responsible. Key ways to improve visibility are:

- Creating a webpage with an overview of the project structure, consortium members, partners, and students involved in the project.
- All progress will be tracked in one place by means of an online project management tool that is supported across the three institutes. This includes a visualization of the progress on the deliverables determined for each work package and publications.
- Published articles are linked on the website with additional information for stakeholders and laypersons.
- Activity report is quarterly published on the website.
- A bimonthly internal newsletter is sent to all active members of the flagship program.
- During the bimonthly update meetings (for all active members), one challenge of the project will be discussed

Externally

Because the topic **CODE RED** is of a high societal relevance, our aim is to provide laymen summaries via Instagram, podcasts or Youtube videos to raise awareness and to reach out to the public actively. Furthermore, since investigation of citizens' and patients' experiences and preferences are part of our program, we will engage patient organizations to cooperate in the research design and conduct of the empirical studies and normative analyses within our project.

Attracting talents

Externally

Based on our extensive network from a variety of academic disciplines we will be a powerhouse to attract talents in the **CODE RED** field, especially those with an interest in both academic healthcare related research and societal responsibility.

Each Work Package includes senior academic staff, early-career researchers, and students, and external stakeholders can be consulted for each multi-disciplinary Work Package.

Internally

The bimonthly newsletter is used to publish internal open calls for PhD students and BA/MA students for research activities. This gives an opportunity to flagship participants to find the right talents for their projects while building a stronger network in which early career researchers and students get to know the full project. In other words, the 'pool' of (PhD) students will be considered a joint asset.

Further, to engage and guide participating students a students' workgroup 'Student Spirit Team' will be created which is managed by the Flagship Student's Coordinator. Following the 'talent attracts new talent' logic, this workgroup will help to attract and involve new students.

Attracting partners and funding

In our extensive and top-notch consortium, it will be easy to attract more partners. Funding possibilities are available on a national level (e.g. ZonMw research programs Dutch *Groeifonds* initiative), international level (Horizon) and via investment fund options. Regarding to the last option, first explorations have been initiated.

2.e Long term societal impact

CODE RED strives for the following long-term contributions to healthcare:

- Sustainable and efficient healthcare
- Tailored drug dosing
- More efficacy and less side effects of drugs
- Improved patient journey with respect for differences in cultural background
- Patient's return to society, both beneficiary to an individual patient as to society
- Patient empowerment and respect for autonomy by enabling patients to make better informed choices about medical treatment
- Enabling doctors to make better informed choices when prescribing medicines, taking cost-effectiveness into account
- Subsequent field-testing of research results as a standard practice – Erasmus MC as a living lab – will contribute to enhance sustainable affordability of innovative drugs and therefore access to medicines for those in need.

Students from different disciplines and faculties are welcome to engage in projects derived from the flagship work packages. Students will be made aware of these projects via the regular sources and department procedures. The last-year developed minor Toxic at the Pharmacy Department of Erasmus MC which is open to students of various disciplines and was a great success and bundled students with different perspectives on major elements of the drug lifecycle.

By adopting the lifecycle approach, we will be able to actually influence common practice derived from our research and field-testing experiments. When in place we make sure to cooperate with our Institutes' Technology Transfer Offices to valorize our findings.

3. Flagship proposal (25%) max 2000 words for 3.a to 3.d

3.a Ambitions

Contribution to excellent Convergence H&T research

CODE RED aims to make an innovative and structural academic contribution to evidence-based knowledge and practices that help to reduce costs across the drug lifecycle, improve patient care and secure patient access, while crossing the boundaries between institutes and disciplines. Envision the contribution of bio-nanoscience to potentially creating synthetic copies of biological cells for, eventually, drug development and production at much lower costs, or the development of algorithms providing optimal decision rules for drug treatment and dosing. Hence, **CODE RED** specifically bundles the strengths of exact sciences with humanities to create comprehensive and ethically responsible solutions to the challenges that expensive drugs present to affordability, access, and quality of care.

Research and approach

Although many studies have shown interest in specific aspects of drugs pricing, cost-effectiveness and personalised treatment, the novelty of **CODE RED** lies in the overall structure which integrates the full drug lifecycle (figure 2). Additionally, the project contributes to technologies that are at the forefront of current research and potentially at the forefront of future practice. Examples from the seven work packages are bio-nanoscience research into potential synthetic copying of biological

cells (WP 1), using innovative statistics and prospective study designs (Real World Data) to accelerate drug (WP2), developing and piloting new pricing, reimbursement and purchasing models (WP 3 & 4), assessing patient and societal preferences regarding drug pricing and switching (WP 5), establishing artificial intelligence to optimize decision making in drug treatment and pricing (WP 5 & 6) and establish cost-effectiveness models (WP7). This combined approach is in line with Convergence Health & Technology because it advocates for crossing boundaries between institutes and disciplines to develop an ecosystem in which grand challenges can be tackled and introduced in common and future practices.

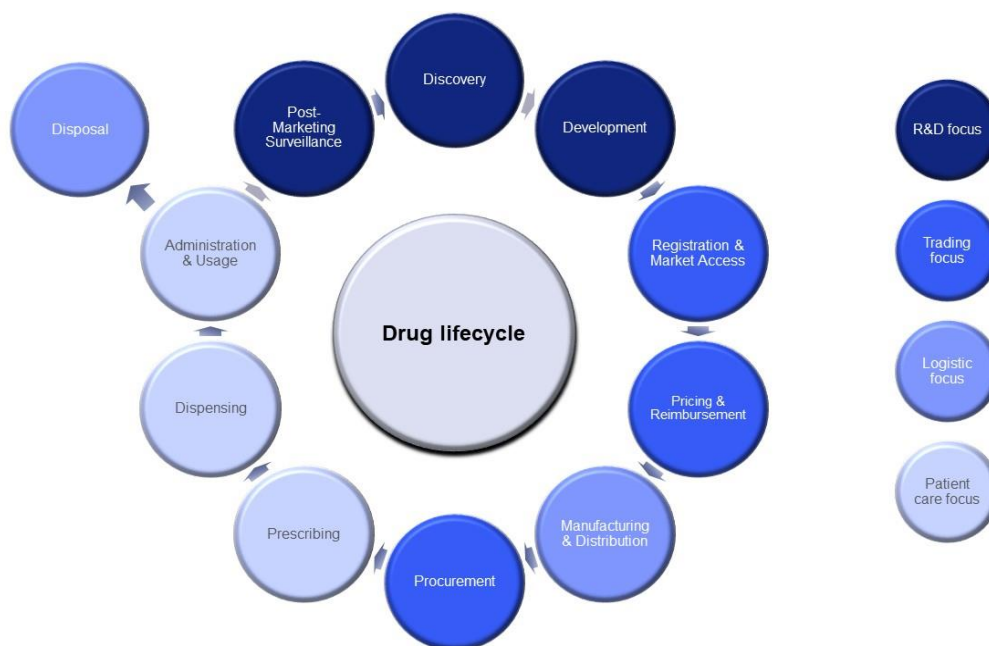


Figure 2. Drug lifecycle and specific focus areas

3.b Aims and feasibility

Aim

To structurally reduce pharmaceutical expenditure across the drug lifecycle and to improve patient care using innovative technology and knowledge convergence.

Objectives

To:

- I. Develop drug cost reduction frameworks for sustainable health care systems.
- II. Gain knowledge and stimulate the application of personalised treatment for patients to improve treatment outcomes and to optimize precision drug dosing.
- III. Promote a data-driven approach in which human-centred artificial intelligence (AI) and machine learning (ML) are applied as tools to ensure transferability of findings
- IV. Establish innovative methods to accelerate registration and subsequent production of innovative drugs.
- V. Create a research environment that stimulates convergence between the different stakeholders and disciplines across the drug lifecycle.

- VI. Ensure long-term sustainability of the knowledge and tools produced by **CODE RED** by adequate impact evaluation and stakeholder engagement throughout and beyond the project.

Student engagement

Within our project we wish to attract students from various disciplines. On one hand, our seven work packages leave room to a variety of students to practice their skills and knowledge and to actively participate in research. On the other hand, we will establish a yearly partly renewing Student Spirit Team which will be invited to critically question all CODE RED staff and research. Moreover, this team will play an important role by setting up innovative ways of communication and cross-pollination within our project as outside.

Feasibility

By establishing an excellent consortium with clear roles and responsibilities (see Section 4), practical implementation of the Work Package is ensured. Since all included key disciplines are well-established within the three institutions, matching can be facilitated through consortium members with relevant networks, experience and opportunities for grant-writing. Additionally, there are sufficient perspectives on the available human resources: each participating institute and department has committed to delivering 'their' share of manpower to safeguard adequate execution of the researches stated in the Work Plan.

A potential risk to the feasibility of the project could be the inclusion of some high-risk-high-gain research in the flagship, particularly the synthetic copying of biological cells in Work Package 1. Nevertheless, we circumvent this risk by integrating studies with low uncertainty on deliverables in the other Work Packages. Hence, the success of **CODE RED** is not dependent on the success of the high-risk-high-gain research, yet it potentially delivers a substantial, innovative contribution to science by including these projects.

3.c Work plan

Work plan

Project structure

In line with the mission-oriented approach displayed in Section 2.b., **CODE RED** involves the following bodies and roles:

- I. Coordinator and program secretary
 - The **CODE RED** coordinator and program secretary are responsible for the visibility and stakeholder engagement as well as progress evaluation of the flagship project.
 - Appointed coordinator: Aniek Dane (Erasmus MC)
- II. Institute leads
 - See Section 4 'Consortium')
- III. Advisory Board
 - Function: The Advisory Board is accessible for advice and input for all **CODE RED** members to enhance decision-making and research output. These stakeholders are either from institutes other than the three Convergence institutes or specialize in specific overarching or practical fields (e.g., negotiation or ethics).
 - Appointed actors:

- Teun van Gelder, professor of clinical pharmacology, LUMC
- Jan-Dietert Brugma, director outpatients' pharmacy, Erasmus MC
- Vincent Suttorp, strategic purchase advisor (medicines), Erasmus MC
- Naomie Tielen & Guido Koster, Ferring
- Department of Sales, Erasmus MC (specific appointed person to be selected)

IV. PhD Work Group and Student Spirit Team

- Function: PhD students and BA/MA students should experience a sense of 'belonging' and 'ownership' within CODE RED so they can be autonomous and creative with their talents. By creating both a PhD and BA/MA student Work Group, knowledge and talent can be shared among each other on a continuous basis.
- Actors:
 - PhD and MA/BA students
 - PhD Coordinator: Anne-Sophie Klein Gebbink (Erasmus MC)
 - Student Coordinator: to be appointed

Work packages

To reduce expenditure on drugs for society, all phases of the drug lifecycle should be (re)assessed. We aim to launch seven intertwined work packages that help to take on a comprehensive approach to drug pricing and expenditure while maintaining and improving quality of care for patients:

Work package 1: Academic innovation	
Key disciplines	Biochemistry, bionanoscience, AI, hospital pharmacy, internal oncology
Key participants	Marileen Dogterom (TU Delft), Christophe Danelon (TU Delft), Hugo van der Kuy (Erasmus MC), Martijn Lolkema (Erasmus MC)
Aim	Conducting translational research into the potential of creating synthetic copies of biological cells and identifying DNA and RNA biomarkers for personalized treatment
Activities	1) Conducting research to assess the basic potential of synthetic copying of biological cells to potentially develop more stable and hopefully cheaper variants of biologics, to begin with synthesizing oncolytic viruses. 2) Conducting both large scale and in-depth data collection for biomarker research for to a) enhance personalised medicine (by predicting the probability of treatment response) and; b) early response evaluation (by enabling earlier detection of treatment effects). 3) Finding new applications for already registered and off-patent oncology drugs (drug repurposing).

Work package 2: Local production & registration	
Key disciplines	Clinical study designs, real world data, clinical pharmacy, radiology, regulatory affairs
Key participants	Jan Cornelissen (Erasmus MC; EMA), Aad van der Lugt (Erasmus MC)
Aim	Exploring the potential and implications and innovative study designs and statistical methods to accelerate drug registration and drug dosing and alternative methods for drug production and marketing authorization by public and/or academic stakeholders.
Activities	1) Rapid and small phase II studies with Real World Data as prospective controls, incorporating innovative statistics to accelerate drug development and validate alternative dosing schemes of recently approved but very expensive drugs e.g., venetoclax combination therapy in elderly AML patients. 2) Introduction of an ICT-based system in cooperation with The Dutch Cancer Registry and the company Performation to monitor Real World Data prospectively and on a real-time basis by using the entire complexity of digital hospital data (the RHONDA initiative). 3) Case study on potential and barriers of local registration (with haematology department and Apotheek A15) 4) Assessment of the potential economic impact of academic or public registration of repurposed drugs 5) Performing research into the potential of local/academic production of radiopharmaceuticals, for example Lutetium-177- PSMA
Work package 3: Alternative pricing & reimbursement	
Key discipline	Health economics
Key participants	Carin Uyl-de Groot (EUR), Frederick Thielen (EUR), Anne-Sophie Klein Gebbink (Erasmus MC)
Aim	To study and develop alternative pricing, payment and reimbursement models that are adequately fit for innovative pharmaceuticals and novel health therapies (e.g., (ultra-) orphan drugs, ATMPs).
Activities	1) To report on insight in practical potential of pre-existing alternative price models by means of literature reviews, document analyses, interviews, and surveys. 2) Evaluating access to new pharmaceuticals by the use of specific (Orphan) Drug Access Protocols ((O)DAP)

	3) Developing a new pricing model which considers an optimal balance between social and economic entrepreneurship. 4) To review / Report on insight in practical potential of pre-existing alternative pricing models 5) Development of new appropriate reimbursement models/payment schemes for innovative pharmaceuticals (including models /schemes based on cost-effectiveness and budget impact). 6) Assessment of impact of new reimbursement models and payment schemes. 7) Updating the master elective “Pharmaceutical Pricing and Market Access” at ESHPM with findings of this work package.	
Work package 4: Purchasing methods		
Key disciplines	Hospital pharmacy, outpatients’ pharmacy, procurement, health economics	
Key participants	Jan-Dietert Brugma (Erasmus MC), Maurits de Jong (Erasmus MC), Vincent Suttorp (Erasmus MC), Frederick Thielen (EUR), researcher institute for Medical Technology Assessment (iMTA) (EUR)	
Aim	Assessing different purchasing methods for expensive drugs as well as potential innovative models to enhance performance in terms of purchasing power.	
Activities	1) Providing a comparison of different purchasing methods and purchasing groups and their performance, including recommendations on potential innovations to enhance the performance of purchasing methods in terms of purchasing power and reimbursement 2) A study assessing the price impact of transference of drugs to the hospital budget (‘overgehevelde geneesmiddelen’) 3) Conducting a study to health insurers' purchasing methods and their influence on the market 4) Teaching medical, pharmacy and health policy students regarding the pharmaceutical playing field (part of the successful minor Toxic, an adaptive and multi-disciplinary program in several faculties from different universities)	
Work package 5: Societal & patient engagement		
Key disciplines	Behavioural science, medical ethics, public health	

Key participants	Eline Bunnik (Erasmus MC), Carin Uyl- de Groot (EUR), Anne-Sophie Klein Gebbink (Erasmus MC)
Aim	Providing guidance to help ensure that new knowledge, tools and methods for drug innovation, registration, production, pricing, reimbursement, purchasing, outcomes, and treatment, aimed at reducing drug expenditures, are in line with ethical requirements and societal preferences.
Activities	1) Conducting a citizens' council – age, gender, socially and culturally diverse – to determine what constitutes a socially accepted fair prices for health therapies such as ATMPs and (ultra-)orphan drugs 2) Performing an empirical, qualitative study on patients' and physicians' experiences and preferences regarding drug switching practices 3) Conducting an ethical analysis of switching practices based on the empirical study. 4) Medical training on cost-effectiveness, joint decision making and switching practices 5) Conducting a study on how raising awareness among patients about drug prices may strengthen patient autonomy by enabling them to make better-informed choices about medical treatments / choices that are more in line with their personal values.
Work package 6: Sustainable & precision dosing	
Key disciplines	Clinical pharmacy, AI, ML, pharmacometrics, sustainability
Key participants	Birgit Koch (Erasmus MC), Ron Mathijssen (Erasmus MC), Nestor Parolya (TU Delft), Asterios Katsifodimos (TU Delft), Roelof van Leeuwen (Erasmus MC), Aad van der Lugt (Erasmus MC)
Aim	This work package aims at systematically reviewing the treatment regimen for expensive drugs through pharmacometric modelling using AI and ML.
Activities	1) Collecting data from clinical practice to create a prediction model of treatment response to prevent underdosing, overdosing or selection on the wrong drugs (using classical statistics and AI). This will be done for nivolumab and pembrolizumab, and e.g., by use of radiomics. 2) Running case studies to test/pilot the performance and practical implications of the developed algorithms (see 1). 3) Studying the optimum dose of pembrolizumab (and also nivolumab, which is comparable to pembrolizumab) in the treatment of patients with solid tumors or lymphomas, with

	use of one of the largest national databases on patients with solid tumors, named MULTOMAB.
Work package 7: Real-world outcomes & cost-effectiveness	
Key disciplines	Medicine, AI, ML, optimizing, health economics
Key participants	Frank Leebeek (Erasmus MC), Nestor Parolya (TU Delft), Asterios Katsifodimos (TU Delft), Carin Uyl-de Groot (EUR)
Aim	To establish (cost-)effectiveness in the real-life situation and to develop and test models and algorithms aimed at optimising real-life effectiveness of expensive therapies. Moreover, to draw connections between cost-effectiveness and pricing and reimbursements schemes. A hospital perspective will be taken as decisions on dosing strategies are usually made locally and are likely to have most impact on the hospital's own budget.
Activities	1) Conducting a study to collect real-world (cost)effectiveness data on selected new expensive drugs in which patients are treated with these drugs will be prospectively followed and prespecified outcomes will be monitored and compared. Intended therapies for this are AAV-based gene therapy, treatment with pembrolizumab, and nivolumab. 2) To develop a transferable algorithm regarding cost-effectiveness and optimal treatment decisions based on the real-world cost-effectiveness data and to ensure that these models can be applied for other diseases and disorders for which (several) new treatments have become available or will be in the years to come. 3) Piloting the practical implication of the obtained cost-effectiveness models of treatment regimens in the case studies through conducting case studies of the algorithm(s).
<u>Project joint activities</u> Next to as described in section 2 all consortium members including students will engage in a bi-yearly round table debate to discuss progress, to get inspired by an external speaker and to work on teambuilding and maintaining excellent relationships. The consortium and all its members will provide an inspiring, hard-working and yet socially safe and inclusive environment. <u>Data and infrastructure</u> We will use the digital research environment (DRE), which is already available in Erasmus MC and EUR. In line with standard regulations, a data management plan will be developed.	

3.d Alignment of the program with relevant national and international research agendas

- The National Scientific Agenda ('Nationale Wetenschapsagenda', NWA) is pointing towards the need for increased interdisciplinary research that connects research and society. Since **CODE RED** uses science and technology as a vehicle to improve health outcomes and reduce costs through personalized medicine, the project is in line with this overarching aim.
- The Nationaal Groeifonds initiatives to foster long-term economic growth by means of investing in research, innovation, infrastructure and knowledge transfer.
- The Federation of Medical Specialists (FMS) has launched its vision on the Medical Specialist of the future in its document Medisch Specialist 2025.

4. Consortium (25%) max 2000 words for 4.a to 4.d**4.a Track record of the consortium members**

1	Prof. dr. Hugo van der Kuy (Flagship lead from Erasmus MC)	<i>Affiliation</i> Head of Department, Full Professor of Hospital Pharmacy, Erasmus MC <i>Academic track record</i> ▫ Research focus on implementing different strategies to improve efficacy, safety and the spending of drugs in general, but expensive drugs in particular. ▫ Teaches several courses both at Erasmus MC and Erasmus University Rotterdam. ▫ Experienced in translating experimental methods into clinical research, e.g., midazolam nasal spray, ventrain, heterologous boosting and many other projects. ▫ Associate editor European Journal of Hospital pharmacy. <i>Student supervision</i> Elaborate experience in engaging & supervising BSc and Msc students: ▫ >10 BA students ▫ >10 MA students ▫ >10 PhD students <i>Funding</i> ▫ Approx. 4M euros total funding (e.g., ZonMw/Senternovem/KNMP)
2	Prof. dr. Carin Uyl- de Groot (Flagship lead from Erasmus University Rotterdam)	<i>Affiliation</i> Full Professor of Health Technology Assessment, Erasmus University Rotterdam Director institute for Medical Technology Assessment (iMTA b.v.) <i>Academic track record</i> ▫ Research focus on cost-effectiveness and outcomes studies.

		▫ Member of the Dutch Health Council since 2015. ▫ PI of several registries and outcomes studies, part of several international projects (e.g., Horizon, National Institute of Health USA). ▫ Reviewer for several high impact journals ▫ Published research in e.g. including Lancet Oncology, Nature Reviews Clinical Oncology, Journal of Clinical Oncology and Blood, and conferences. <i>Student supervision</i> ▫ 11 MA ▫ 23 postdocs/researchers ▫ 15 PhD <i>Funding</i> ▫ > 1M euros total funding ▫ Relevant for CODE RED: • Personal grant: 2nd round ERC advanced grant 2021 (result May 2022) • Other (potential) grants: AICCELERATE (HORIZON 2020-2023, € 750.000), T2EVOLVE (IMI EUROPE, 3years, € 170.000) (2021-2024), AIO financed by IKLN, Holland PTC (1 AIO) CADTH project, adaptation disease model multiple myeloma (€ 70.000) (2021-2022), SKMS Ned Internisten Vereniging (Euro 54,000) (2021-2022).
3	Prof. dr. A.M. (Marileen) Dogterom (Flagship lead from TU Delft)	<i>Affiliation</i> Full Professor in bionanoscience, TU Delft Medical Delta Professor at the Leiden Institute of Physics <i>Academic track record</i> ▫ Internationally renowned expert in experimental cell biophysics with a pioneering track record in biophysical research of the microtubule cytoskeleton ▫ Published ~100 papers in high-ranking physics and biology journals ▫ Co-initiator of the European Synthetic Cell Initiative. ▫ Received the Dutch Physica Prize in 2015, Spinoza award in 2018 ▫ Board Member, of the Royal Netherlands Academy of Arts and Sciences (KNAW). <i>Student supervision</i> ▫ 17 alumni Currently: ▫ 9 BA/MA ▫ 7 PhD/postdocs

		<i>Funding</i> Extensive possibilities for funding for 'own' Work Package (WP 1) (exact funding TBD) Please note that the Letter of Commitment of Prof. dr. Dogterom is currently missing. Nevertheless, we have received verbal consent.
Consortium members per work package (alphabetical order):		
4	Jan-Dietert Brugma MSc	<i>Affiliation</i> Director Outpatient Pharmacy, Erasmus MC <i>Academic track record</i> - Pharmacist - Specialized in the drug reimbursement issues - Founded the outpatient pharmacy of Erasmus MC (2012) - Member of the Round Table Orphan Drugs ('Ronde tafel Weesgeneesmiddelen'), chaired by the Dutch Health Care Institute <i>Student supervision</i> N.A. <i>Funding</i> N.A.
5	Eline Bunnik PhD	<i>Affiliation</i> Assistant Professor, Department of Medical Ethics, Philosophy and History of Medicine, Erasmus MC <i>Academic track record</i> - Expertise in assessing the ethical implications of biotechnological innovations for health care and society - Works with the Rathenau Instituut and NEMO Kennislink to conduct societal discussions on the animal as a source of donor organs in a project funded by VWS - Completed a PhD on Ethical Issues in Genomic Testing at Erasmus MC (2014) - Teaches medical ethics to BA and MA students and healthcare professionals <i>Student supervision</i> 2 PhD alumni Currently: 5 PhD >30 MA/interns

		<i>Funding</i> ▫ > 2 M euros total received (e.g., NWO, ZonMw, KWF, EU Horizon2020) ▫ Currently KWF Dutch Cancer Society grant available (480.000 euros) for use
6	Prof. dr. J.J. (Jan) Cornelissen	<i>Affiliation</i> Internist-Hematologist, Department of Hematology, Erasmus MC <i>Academic track record</i> ▫ Specialized in patients with acute leukemia ▫ Program lead of haemopoietic stem cell transplantations of the Erasmus MC Cancer Institute for the Department of Hematology. ▫ Chair of the Board of the Dutch-Belgian HOVON study group ▫ Principal Investigator of several cell transplant studies ▫ Completed BROK course <i>Student supervision</i> ▫ Total of >25 students (BA/MA/PhD) <i>Funding</i> ▫ > 10 M euros received in total funding
7	Christopher Danelon PhD	<i>Affiliation</i> Associate Professor, Department of Bionanoscience, TU Delft <i>Academic Track Record</i> ▫ <i>Research focus</i> on syntehtic biology, aimed at reconstituting essential cellular modules that will ultimately work in concert to support autonomous growth and division cycles. ▫ PhD degree in membrane biophysics from the University Paul Sabatier in Toulouse in 2003 ▫ Postdoc at EPFL and University of Lausanne, Switzerland ▫ Director of the Casimir Research School at the Kavli Institute of Nanoscience in Delft and at the Leiden Institute of Physics (LION) <i>Student supervision</i> ▫ 4 PhD s ▫ Several BA/MA students within the 'Danelon Lab', a TU Delft-based research group led by Christopher Danelon <i>Funding</i> ▫ Experience with funding and grants, quantification missing
8	Asterios Katsifodimos PhD	<i>Affiliation</i> Assistant Professor, Web Informations Systems group, TU Delft

		<i>Academic track record</i> ▫ PhD INRIA Saclay and the University of Paris-Sud (2013) ▫ Postdoc Database System Group in TU Berlin. ▫ Expertise and teaching in scalable data management; data management and analysis, stream processing, systems for scalable AI/ML, and data integration ▫ Member of the Web Information Systems (WIS) group <i>Student supervision</i> Currently supervises; • 2 postdocs • 5 PhD students <i>Funding</i> ▫ Experience with attracting funding (amount unknown)
9	Prof dr. BCP (Birgit) Koch	<i>Affiliation</i> Hospital Pharmacist-Clinical Pharmacologist, Hospital Pharmacy, Erasmus MC Full Professor of Clinical Pharmacometrics, Erasmus MC <i>Academic track record</i> ▫ Specialized in pharmacometrics and toxicology in special patient populations. ▫ Member executive board of EPASG, the PK/PD group of ESCMID ▫ Member of SWAB committee and the medical ethical board. ▫ Authored > 150 publications on PK/PD, TDM and toxicology ▫ Finished BROK and BKO courses <i>Student supervision</i> N.A. <i>Funding</i> ▫ Received ZonMw grant (x3) ▫ >2M euros total received grants
10	Prof. Dr. Frank W.G. Leebeek	<i>Affiliation</i> Head Department of Hematology, Full Professor in Hematology, Erasmus MC <i>Academic track record</i> ▫ Research interests of Hemostasis and Thrombosis ▫ Completed two BROK courses (Erasmus MC) ▫ Chairman of 'Thema Daniel' for Erasmus MC, the Scientific Advisory Board of Hematon Research Foundation ▫ Board member of European Hematology Association EHA ▫ Heimburger Award, and Swammerdam Prize by the Dutch Hematology Society

		<i>Student supervision</i> ▫ 10 MA ▫ 30 PhD <i>Funding</i> ▫ TTO NWO grant (co—PI), Dutch Heart Foundation, Dutch Haemophilia Foundation ▫ Received > 1M euros total Funding
11	R.W.F. (Roelof) van Leeuwen PhD	<i>Affiliation</i> Researcher and Assistant Professor, Departments of Hospital Pharmacy and Oncology, Erasmus MC, the Netherlands. <i>Academic track record</i> ▫ PhD student on “Drug-drug Interactions in patients treated with anti-cancer agents” (2016) ▫ Postdoc in translational pharmacology at the Department of Medical Oncology (2016-2019) ▫ Research focus on chemo-efficiency with a primary focus on pharmaco-economics ▫ published several peer reviewed papers in clinical journals like Lancet Oncology and the Journal of Clinical Oncology ▫ Lecturer at several (university) teaching programs <i>Student supervision</i> ▫ Supervising several PhD students, qualification missing <i>Funding</i> ▫ <i>Non profit:</i> Coolsingel Stichting; Stichting Mitialto; Merel Stichting. ▫ <i>Other:</i> Astellas; Bayer Pharma; Boehringer Ingelheim; Roche; BMS.
12	M.P.J.K. (Martijn) Lolkema, MD PhD	<i>Affiliation</i> Principal Investigator, Department of Medical Oncology, Erasmus MC <i>Academic track record</i> ▫ Focus area is large scale genomic analyses of metastatic cancer tissues, translational biomarker studies in urogenital cancers and early phase drug development in oncology. ▫ Head of the phase I unit of the Erasmus MC Cancer Institute in Rotterdam ▫ Course coordinator and lecturer on clinical trial development, minor oncology, and clinical cancer research. ▫ Completed BROK course <i>Student supervision</i> ▫ 6 PhD students

		<i>Funding</i> ▫ > 1 M euros total grants received (e.g., Alpe d'Huzes/ KWF, project #7080, KWF (EMCR 2015-8037), NOW)
13	Prof. dr. A. (Aad) van der Lugt	<i>Affiliation</i> Radiologist, Director of Stroke Center, Professor in Neuroradiology and head/neck radiology, Head Department of Radiology, Erasmus MC <i>Academic track record</i> ▫ Member of the Research Committee of the European Society of Radiology ▫ Chairman of the Working Group on Personalized Medicine (2013-2015) ▫ Present chairman European Imaging Biomarkers Alliance (EIBALL) <i>Student supervision</i> ▫ 10 BA ▫ 10 MA ▫ 28 PhD <i>Funding</i> ▫ 7.5 M euros total grants received
14	Prof. dr. A.H.J. (Ron) Mathijssen	<i>Affiliation</i> Internist-Oncologist, Clinical Pharmacologist Department of Internal Oncology, Full Professor in Individualized Oncological Pharmacotherapy, Erasmus MC <i>Academic track record</i> ▫ Head of the Laboratory 'Translational Pharmacology' and workgroup 'Personalized Medicine' ▫ Chair of Dutch Pharmacology Oncology Group ▫ Committee member multiple societies outside Erasmus MC: e.g., KWF, Oncode, NIV, NVKF&B. ▫ Head section Translational Pharmacology ▫ Completed BROK course <i>Student supervision</i> ▫ 48 MA ▫ 35 PhD <i>Funding</i> ▫ > 10 M euros total grants received (Dutch Cancer Society, ZonMw, pharmacy etc).

15	Nestor Parolya PhD	Affiliation Assistant Professor in Statistics at Delft Institute of Applied Mathematics, TU Delft Academic track record ▫ Research focus on High-Dimensional Statistics and Random Matrix Theory, Mathematical and Statistical Finance, and Financial Engineering and Operations Research ▫ PhD in Economics ▫ Coordinator of “Minor Finance”, Department of Applied Mathematics, TU Delft ▫ Wolfgang Wetzel Award of the German Statistical Society (2019) Student supervision Currently: ▫ 4 BA ▫ 4 MA ▫ 2 PhD Funding ▫ Internal grants awarded TU Delft (quantification missing)
16	Prof. dr. M.H.N. (Maartje) Schermer	Affiliation Professor in Philosophy of Medicine, Department Medical Ethics, Philosophy and History of Medicine, Erasmus MC Academic track record ▫ Expertise in bioethics, ethics of screening and early diagnostics and philosophy of medicine. ▫ Teaches and coordinates the Master research Ethics & Philosophy as well as various post-graduate courses. ▫ Chair of the standing committee on Ethics and Law of the Health Council of the Netherlands and of the Centre for Ethics and Health (CEG) Student supervision ▫ Extensive experience with supervising students, quantification missing Funding ▫ >1 M euros total grants received
17	Frederick Thielen PhD	Affiliation Assistant professor, Erasmus School of Health Policy and Management (ESHPM), Erasmus University Rotterdam Academic track record

		▫ PhD on 'Current Challenges in Health Technology Assessment: Assessing the costs and cost-effectiveness of novel treatment in haemato-oncology' (2020) ▫ Research focus on economic evaluations, cost, and outcome studies. ▫ Teacher and co-coordinator MA courses health economics & HTA <i>Student supervision</i> Currently supervises: ▫ 4 MA students ▫ 1 PhD student <i>Funding</i> ▫ > 250.000 euros total grants received ▫ Examples: H2020/Horizon Europe: RETHRIM, T2Evolve
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4.b Expertise of the consortium members in relation to the aims and objectives of the program

As indicated in the above track record, all consortium members are authorities in the field **CODE RED** covers. Their joint expertise will lead to synergy and holds a promise to deliver on our aim.

4.c Involvement of partners

All leads have already existing external ecosystems which consists of a variety of public and private partners, in particular patient organizations (e.g. KWF), health insurers (e.g. CZ, ZN), EMA, pharmaceutical companies (e.g. Ferring), Apotheek A15, HOVON and Performance. Some of these organizations are involved in **CODE RED** via funding, some via participating in our advisory board and some are involved in the studies.

4.d Governance, organization and collaboration strategy of the consortium

CODE RED will be led by the main leads who will be supported by a program secretary and a supervising coordinator. Work package leads are responsible for conducting studies and supervision of their PhDs and postdocs. Meetings will be arranged on a regular base with the senior staff (main leads and WP lead) as well as with all **CODE RED** members. Students of the Student Spirit Team will be challenged to propose and execute creative ways to stimulate cross-pollination and to help establish a vibrant **CODE RED** community culture. Finally, by installing an advisory board connecting to the world outside the academia is established.

Annexes (do not count toward maximum number of pages)

I Budget table (including the external funding table)

See attachment (excel)

II Letters of intent and letters of commitment

See attachments

III Letter of support from department head of the main lead

See attachment

IV References

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